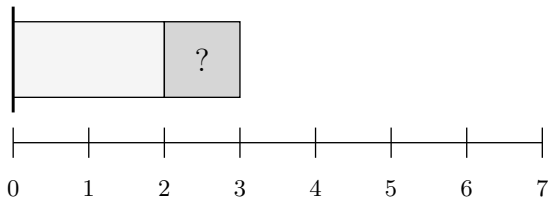
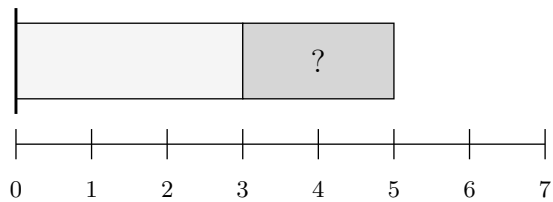


A hidden end can be anywhere in its gray band, including the edges. Each trial may use new lengths. Join strips end to end from 0, with no gap or overlap.

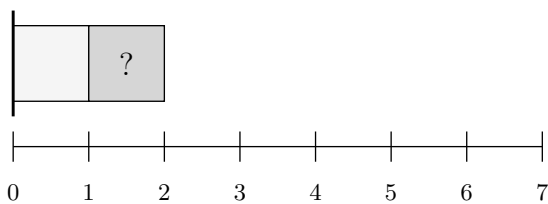
A: 2 to 3 units



A + B: 3 to 5 units

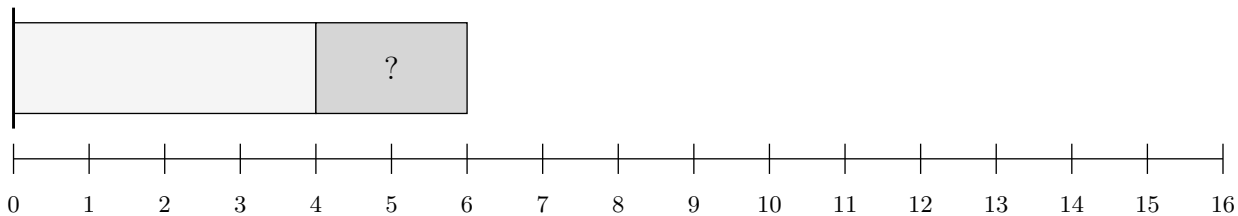


B: 1 to 2 units

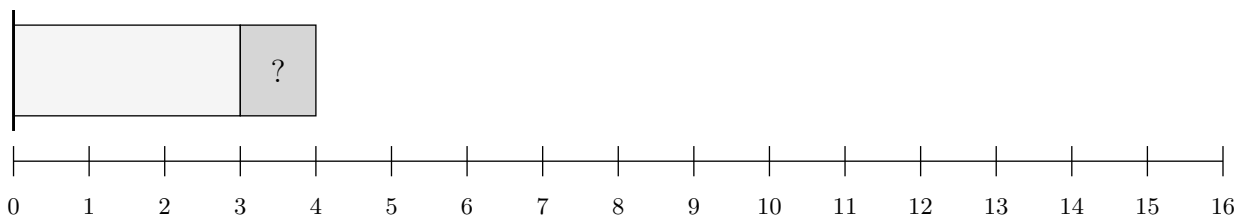


Problem 1: Find the tightest guaranteed range for the join. Someone claims it must end between 8 and 9; build allowed strips that defeat that claim.

A: 4 to 6 units

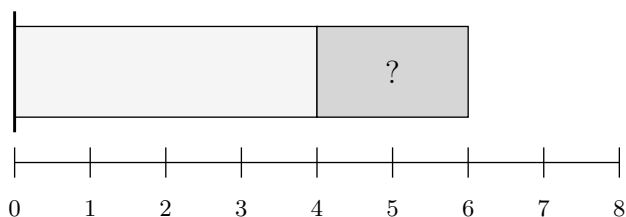


B: 3 to 4 units

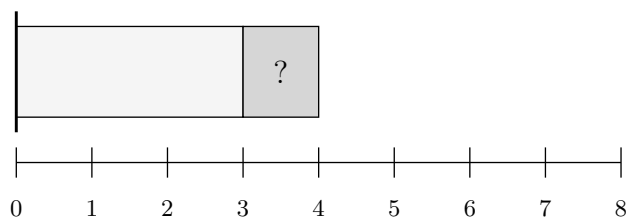


Problem 2: Try only A with B, C with D, and E with F. Choose a pair whose join always reaches 8 and fits within 10. For each rejected pair, let a partner defeat one promise with allowed lengths.

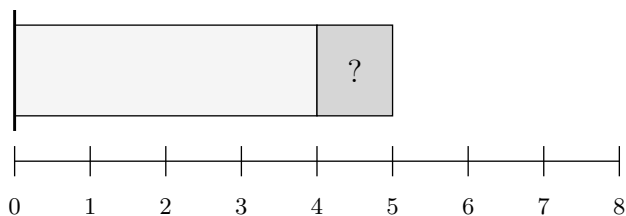
A: 4 to 6 units



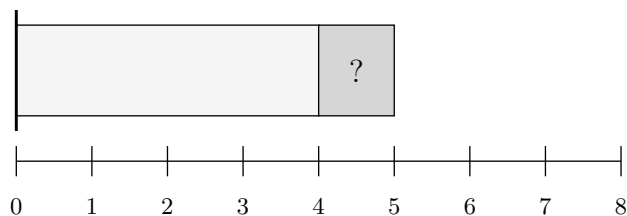
B: 3 to 4 units



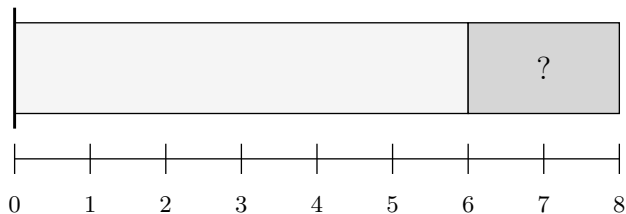
C: 4 to 5 units



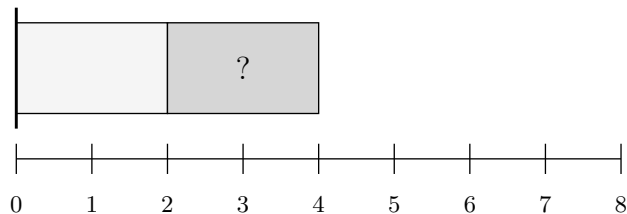
D: 4 to 5 units



E: 6 to 8 units



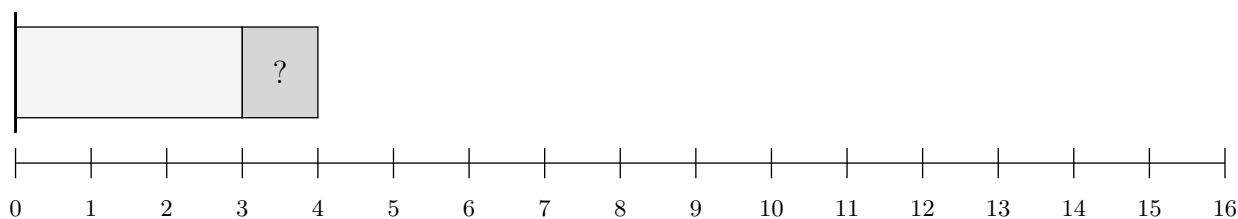
F: 2 to 4 units



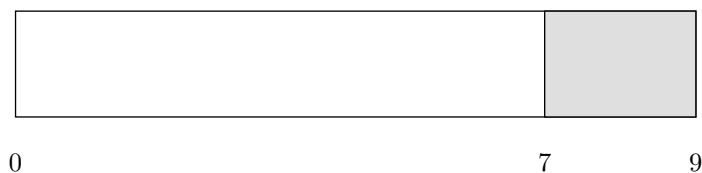
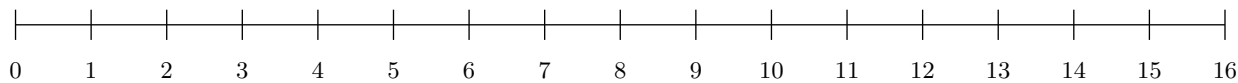
Reaching 8 includes ending at 8. Fitting within 10 includes ending at 10.

Problem 3: A is 3 to 4 units long. Make a range card for B so every join reaches 7 and fits within 9. Make B's allowed range as wide as possible, and explain why it cannot be wider.

A: 3 to 4 units



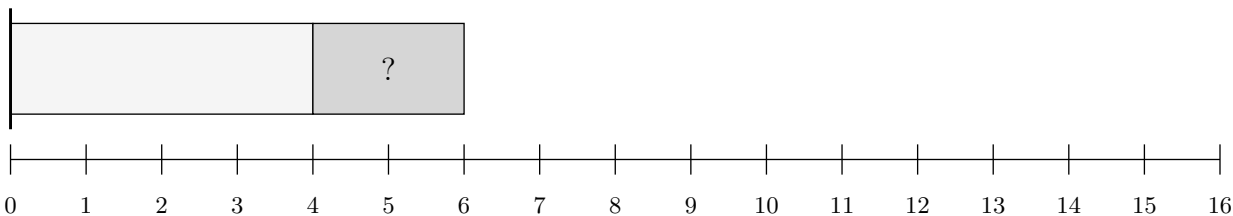
B:



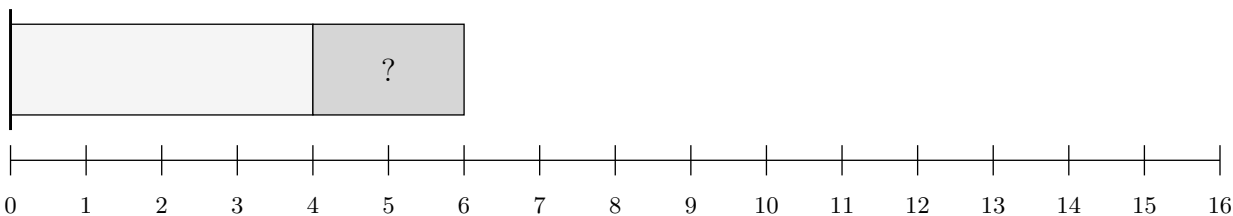
Problem 4: A different pair has A from 4 to 6 and B from 3 to 4. You may learn one exact length. Which strip would you uncover to leave the narrowest guaranteed range for the join?

Problem 5: Compare the gaps between the right ends of these joins, starting at 0. Repeated A means the same unchanged length; B may vary separately. Find the tightest range for each gap.

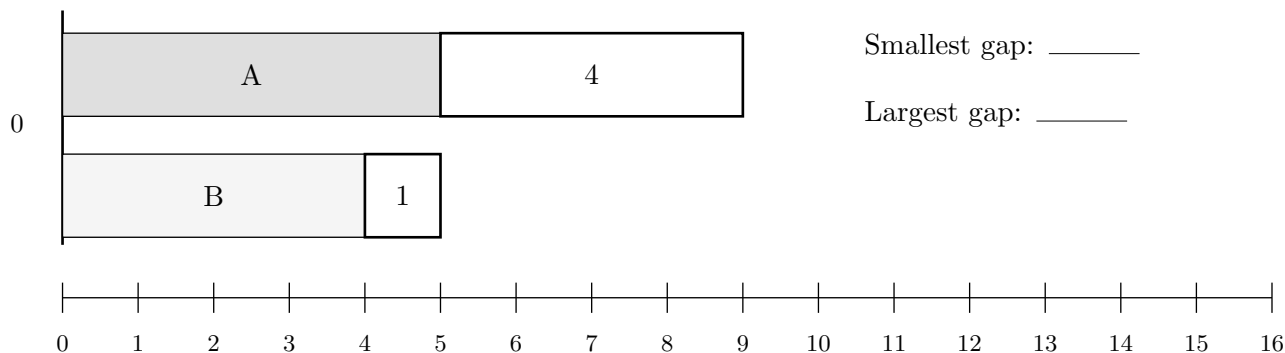
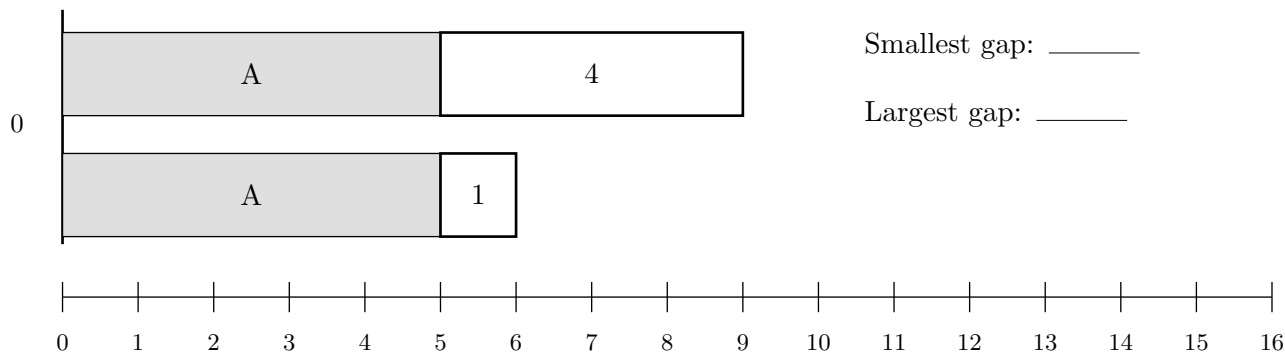
A: 4 to 6 units



B: 4 to 6 units

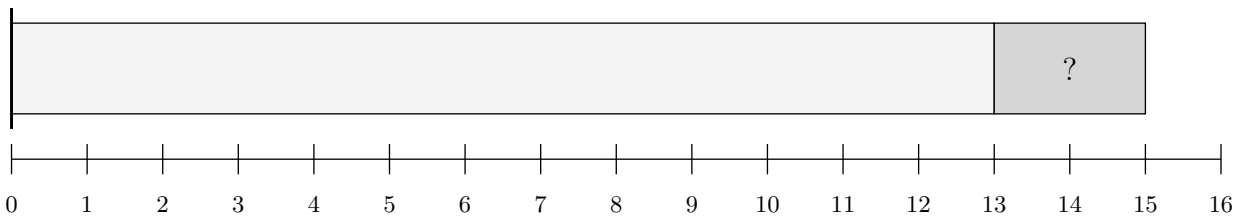


One possible setting; the white pieces are exactly 4 and 1 units long.

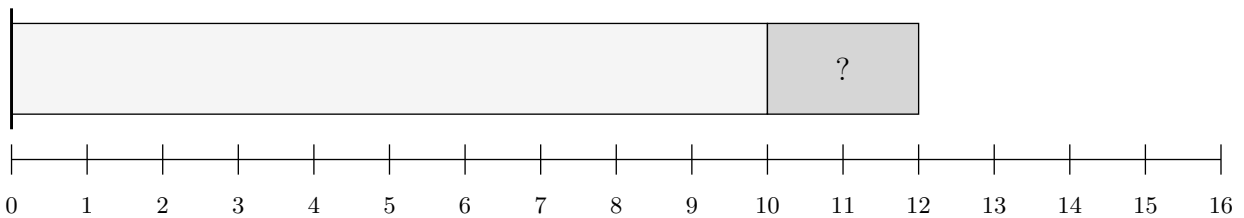


Problem 6: Align both starts at 0. Find the tightest guaranteed range for the gap between the ends. Build lengths that show neither bound can be improved.

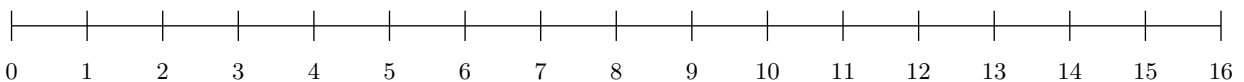
long strip: 13 to 15 units



short strip: 10 to 12 units

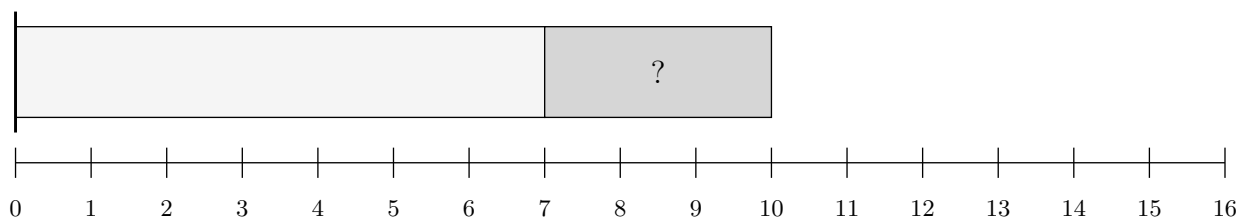


Problem 7: Someone subtracts the two middle values and announces one exact gap. Can you choose allowed strips that show why this is unsafe?

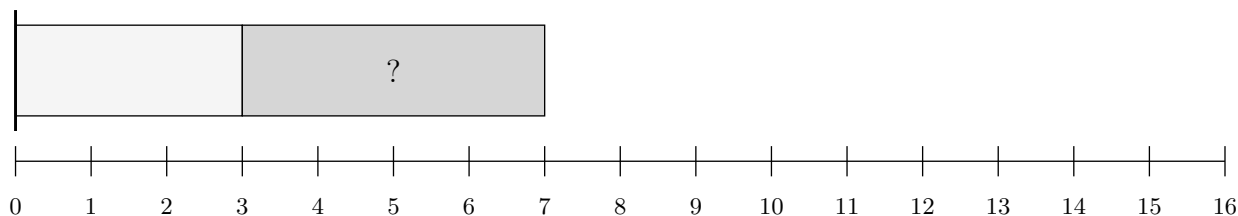


Problem 8: A is 7 to 10 units and B is 3 to 7 units. You learn that their join is exactly 12 units. Align their starts at 0. Find the smallest and largest possible gap between their ends.

A: 7 to 10 units



B: 3 to 7 units



$$A + B = 12 \text{ units}$$

Problem 9: Join A (4 to 6 units) to B (3 to 4 units), then remove that very same A. Someone uses the join's range and A's range separately to claim the remainder could be 1 to 6 units. Is that the tightest range? Explain with the pieces.